

Math708 – Foundations of Computational Mathematics I

Fall 2012

Final Exam

You must show all of your work to get a credit for a correct answer.

1. Given $a \leq x_0 < x_1 < \cdots < x_n \leq b$, let $f(x)$ be a sufficiently smooth function and $H_{2n+1}(x)$ be its Hermite interpolating polynomial of degree $\leq 2n + 1$ satisfying $H_{2n+1}(x_k) = f(x_k)$ and $H'_{2n+1}(x_k) = f'(x_k)$ for $k = 0, 1, \dots, n$. Prove that for all $x \in (a, b)$, there exists a $\eta \in (a, b)$ such that

$$f(x) - H_{2n+1}(x) = \frac{f^{(2n+2)}(\eta)}{(2n+2)!} \omega^2(x)$$

where $\omega(x) = (x - x_0)(x - x_1) \cdots (x - x_n)$.

2. Compute the coefficients and weights $\{(\lambda_j, \sigma_j)\}_{j=1}^n$ of the Gaussian quadrature formulas $\mathcal{I}_n(f) = \sum_{j=1}^n \sigma_j f(\lambda_j)$ for the approximation of $\int_a^b f(x) \rho(x) dx$ where $a = -1$, $b = 1$, $n = 2$ and

$$\rho(x) = \begin{cases} 2, & 0 < x \leq 1, \\ 1, & -1 \leq x \leq 0. \end{cases}$$

3. Let f be a continuous function on $[-1, 1]$ and $\mathcal{I}_n(f)$ denote the Gaussian quadrature formulas $\mathcal{I}_n(f) = \sum_{j=1}^n \sigma_j f(\lambda_j)$ for the approximation of $\mathcal{I}(f) = \int_{-1}^1 f(x) dx$. Prove that

$$\lim_{n \rightarrow +\infty} \mathcal{I}_n(f) = \mathcal{I}(f).$$

4. Consider the family of linear multi-step methods

$$u_{n+1} = u_n + \frac{h}{2} [2(1 - \alpha)f_{n+1} + 3\alpha f_n - \alpha f_{n-1}]$$

where α is a real parameter.

- (a) Analyze consistency and order of the methods as functions of α , determine the value α^* for which the resulting method has maximal order.
- (b) Study the zero-stability of the method with $\alpha = \alpha^*$, write its characteristic polynomial $p(r; h\lambda)$ and, draw and print out its region of absolute stability in the complex plane using computer.